Graph-Based Dependency Parsing

Dependency Parsing: Main Approaches

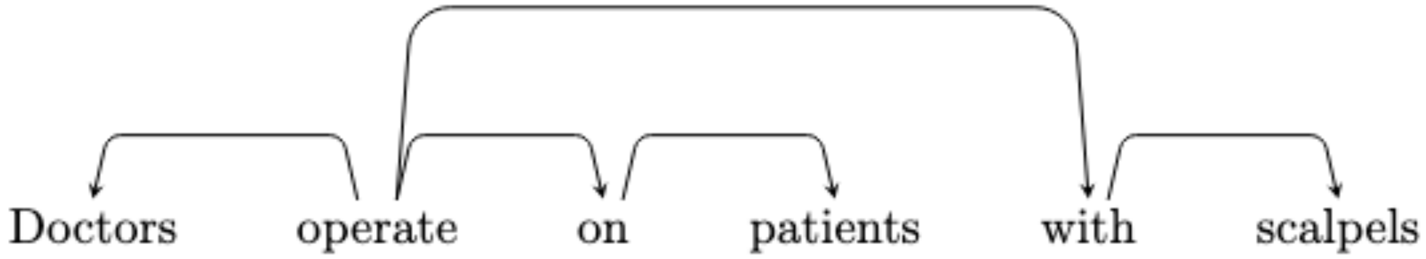
- Dynamic programming:
  - Based on context-free grammars
- Transition-based algorithms:
  - Greedy
  - Left-to-right traversal of the text, each choice is done with a classifier
- Graph algorithms:
  - Structured prediction
  - The sentence is given in advance

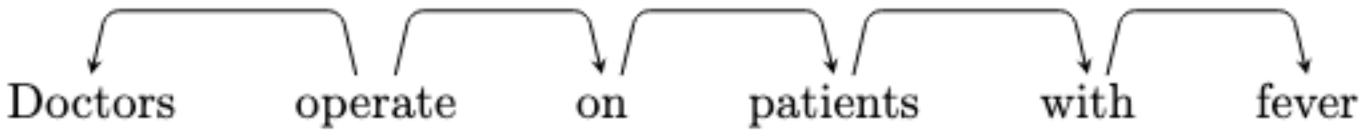
Covered in this course

Dependency Parsing

- What are the sources of information for dependency parsing?
  - Bi-lexical affinities:

    [operate → Doctors] is plausible, [patients → scalpels] less so





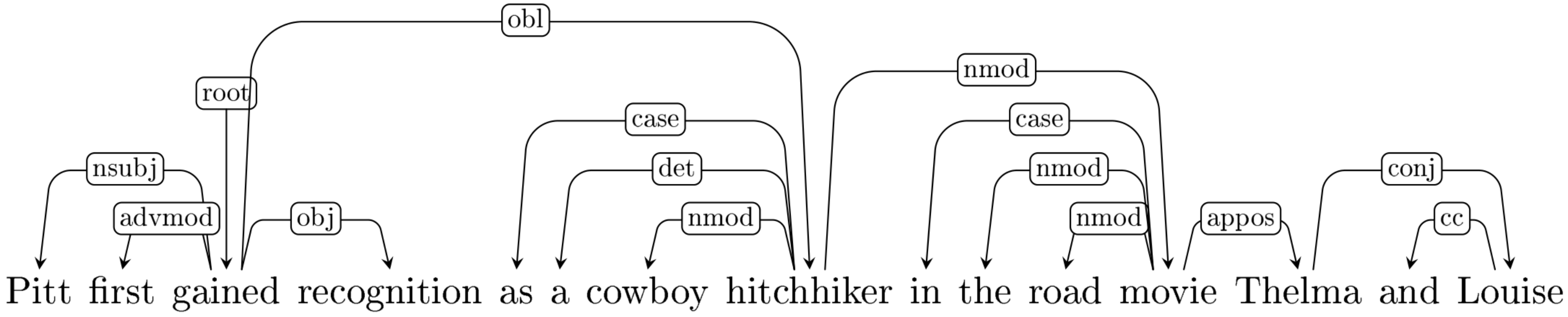
Dependency Parsing

- What are the sources of information for dependency parsing?
  - Bi-lexical affinities:

    [operate → Doctors] is plausible, [patients → scalpels] less so
  - Dependency distance:

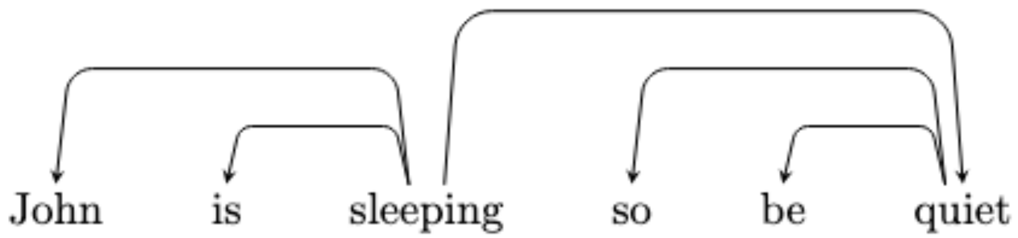
    dependencies are more likely to be with nearby words
  - Intervening tokens:

    dependencies rarely span intervening verbs or punctuation



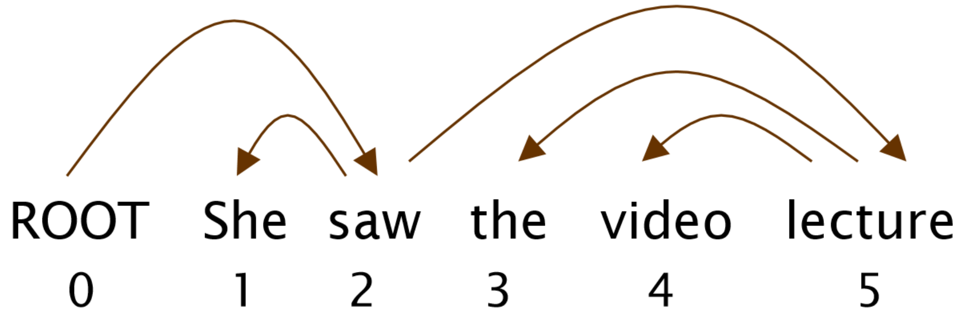
Dependency Parsing

- What are the sources of information for dependency parsing?
  - Valency: how many dependents on which side are usual for a head?
    - [gained] usually has a subject and an object
    - [sleep] does not have an object



    - [give] often has two objects (e.g., ``give **John** **a present**'')

Dependency Parsing Evaluation



$$ACC = \frac{\#CORRECT\ EDGES}{\#NUMBER\ OF\ WORDS}$$

- Accuracy (aka Attachment Score)
- There is Unlabeled Attachment Score and Labeled Attachment Score

Gold

1	2	She	nsubj
2	0	saw	root
3	5	the	det
4	5	video	nn
5	2	lecture	dobj

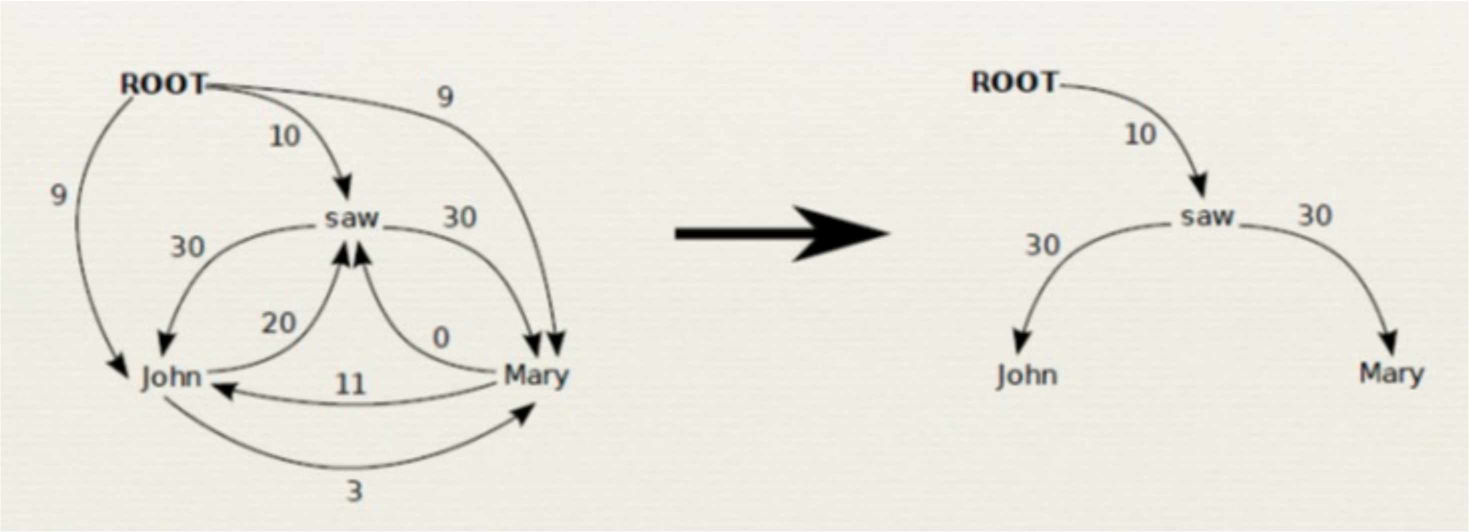
Parsed

1	2	She	nsubj
2	0	saw	root
3	4	the	det
4	5	video	nsubj
5	2	lecture	ccomp

index head index word edge label

Graph-based Parsing

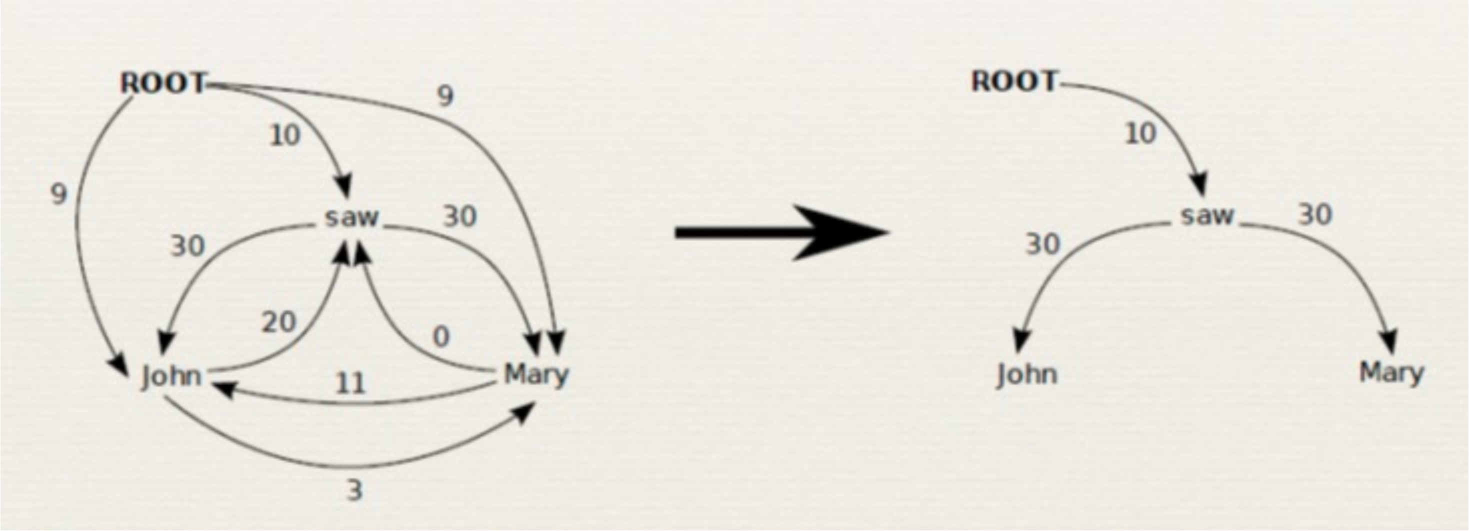
- Graph-based parsing addresses it as a structured prediction problem
- MST Parser:
  1. Score the arcs independently, based on how likely they are to appear in a parse
  2. Find the maximum directed spanning tree over the resulting weighted graph



Online Large-Margin Training of Dependency Parsers
R. McDonald, K. Crammer, and F. Pereira, *ACL 2005*

Graph-based Parsing

- It is also possible to have a multi-graph with several possible edges between each two nodes
  - Useful when using labeled dependency parsing – an edge is needed for each possible labels
  - During inference, it is possible to prune away all edges but the maximal, reducing the multi-graph to a graph



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MST Parser

Define a scoring function over all possible directed trees over $V = \{w_1, ..., w_n, ROOT\}$ where $ROOT$ is the root of the tree. Let $\Phi : V^2 \times L \times S \to \{0,1\}^d$, where L is the label set and S is the set of sentences (feature values can also be real numbers if needed), be a feature function over possible edges.

Let θ be the weight vector (the parameters of the model):

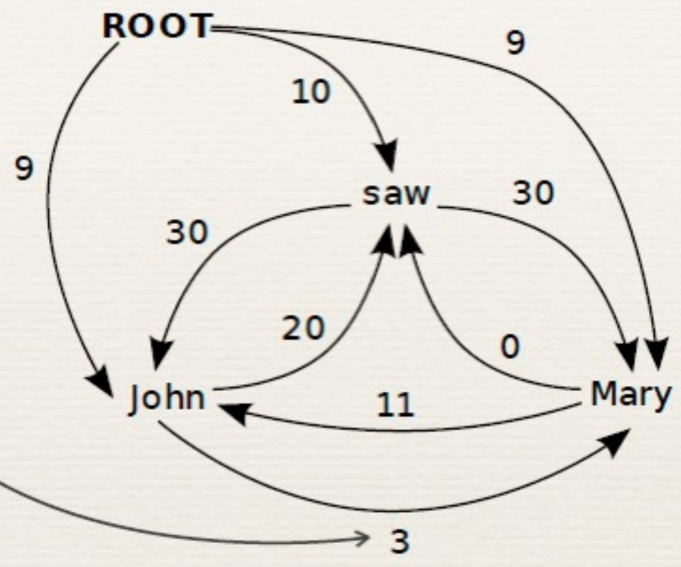
$$score_\theta(v_1, v_2, l \,|x_{1:n}) = \theta^t \cdot \Phi(v_1, v_2, l, x_{1:n})$$

For a directed tree T define:

$$score_\theta(T \,|x_{1:n}) = \sum_{(v_1,v_2,l)\in T} score_\theta(v_1, v_2, l \,|x_{1:n})$$

MST Parser

$$\text{score}(\text{John}\rightarrow\text{Mary}) = \mathbf{w}\cdot\boxed{\mathbf{f}(\text{John}\rightarrow\text{Mary})}$$



Binary Features

Head-POS=NOUN
Mod-POS=NOUN
Head-Word=John
Mod-Word=Mary
In-Between=VERB
Distance=2
Direction=Right
etc.

+conjunctions

MST Parser: Inference and Learning

- Note that inference is simply finding the maximum directed spanning tree
  - We can score each edge based on its features
  - This is done by the Chu-Liu Edmonds algorithm
- It is possible to define this model as log-linear:

$$Pr(T) = \frac{exp(\sum_{(v_1,v_2,l)\in T} \theta^t \cdot \Phi(v_1, v_2, l))}{Z(V, \theta)}$$

- As usual, the gradient of the log-likelihood is given by:

$$\frac{\partial LL}{\partial \theta} = \sum_{i=1}^{N} \Big[\sum_{(v_1,v_2,l)\in T_i} \Phi(v_1, v_2, l) - \mathbf{E}_T\big(\sum_{(v_1,v_2,l)\in T} \Phi(v_1, v_2, l)\big)\Big]$$

MST Parser: Inference and Learning

$$\frac{\partial LL}{\partial \theta} = \sum_{i=1}^{N} \Big[\sum_{(v_1,v_2,l)\in T_i} \Phi(v_1, v_2, l) - \mathbf{E}_T \big(\sum_{(v_1,v_2,l)\in T} \Phi(v_1, v_2, l) \big) \Big]$$

- It is possible to compute the second term exactly, but the algorithm is not simple
- The Averaged Perceptron algorithm provides a simple and useful alternative, by replacing the expectation with a maximum

Structured Prediction Models via the Matrix-Tree Theorem
Koo et al., 2007

MST Parser: Inference and Learning

1. $\theta^{(0)} \leftarrow 0$
2. **for** $r = 1 \ldots N_{iterations}$
3. **for** $i = 1 \ldots N$
4. $T' \leftarrow argmax_T \sum_{(v_1,v_2,l) \in T} score_\theta(v_1, v_2, l)$
5. $\theta^{((r-1)N+i)} \leftarrow \theta^{((r-1)N+i-1)} + \eta \cdot \left(\sum_{(v_1,v_2,l) \in T_i} \Phi(v_1, v_2, l) - \sum_{(v_1,v_2,l) \in T'} \Phi(v_1, v_2, l) \right)$
6. **return** $\frac{1}{N \cdot N_{iterations}} \sum_k \theta^{(k)}$

T_i is the gold standard tree

Learning Rate

Feature function of T' instead of expectation over all possible trees

Discriminative Training Methods for Hidden Markov Models: Theory and Experiments with Perceptron Algorithms
Collins, 2002

Features Used in Graph-based Dep Parsing

John	saw	Mary	McGuire	yesterday	with	his	telescope
N	V	N	N	R	P	PR	N

Features from McDonald et al.

- Identities of the words w_i and w_j and the label l_k

head=saw & dependent=with

Features Used in Graph-based Dep Parsing

John saw Mary McGuire yesterday with his telescope

N V N N R P PR N

Features from McDonald et al.

- Part-of-speech tags of the words w_i and w_j and the label l_k

head-pos=Verb & dependent-pos=Preposition

Features Used in Graph-based Dep Parsing

John saw Mary McGuire yesterday with his telescope

N V N N R P PR N

Features from McDonald et al.

- Part-of-speech of words surrounding and between w_i and w_j

inbetween-pos=Noun

inbetween-pos=Adverb

dependent-pos-right=Pronoun

head-pos-left=Noun

...

Again conjoined with the label (omitted from now on for brevity)

Features Used in Graph-based Dep Parsing

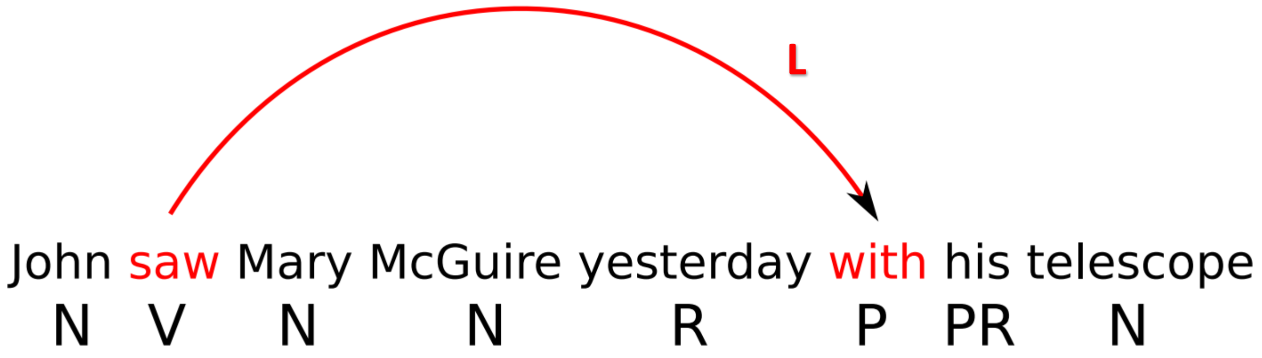
John saw Mary McGuire yesterday with his telescope
N V N N R P PR N

Features from McDonald et al.

- Number of words between w_i and w_j, and their orientation

arc-distance=3
arc-direction=right

Features Used in Graph-based Dep Parsing



Label features

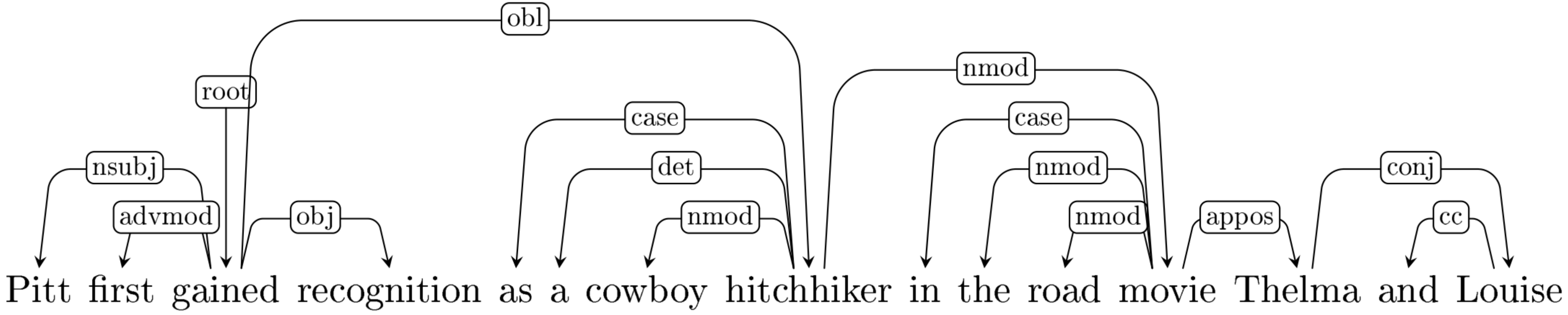
arc-label= **L**

And combinations of all these features…

Graph Based Parsing – Higher Order Features

- MST inference works well when the model factorizes over edges

- However, it has been proven useful to add features that involve several edges, which renders inference intractable
  - These are called “higher order features”



Higher-order Parsing Models

- Higher-order parsing models don't assume that the score of a tree factorizes over edges, but over sets of edges
- For instance, the "grandchild model" assumes that

$$score(T) = \sum_{(i,j,k) \in V^3 | (i,j) \in T \& (j,k) \in T} score(i,j,k) + \sum_{(i,j) \in T} score(i,j)$$

$$score(i,j,k) = \theta_1^t \cdot \phi_1(i,j,k,x_{1:n})$$

$$score(i,j) = \theta_2^t \cdot \phi_2(i,j,x_{1:n})$$

  - We add the grandchild features to the standard MST ones terms (second summation)

Higher-order Parsing Models

- The sibling model with labels would look like this:

$$score(T) = \sum_{(i,j,k)\in V^3 | (i,j)\in T \& (i,k)\in T} score(i,j,k) + \sum_{(i,j)\in T} score(i,j)$$

$$score(i,j,k) = \theta_1^t \cdot \phi_1(i,j,k,x_{1:n})$$

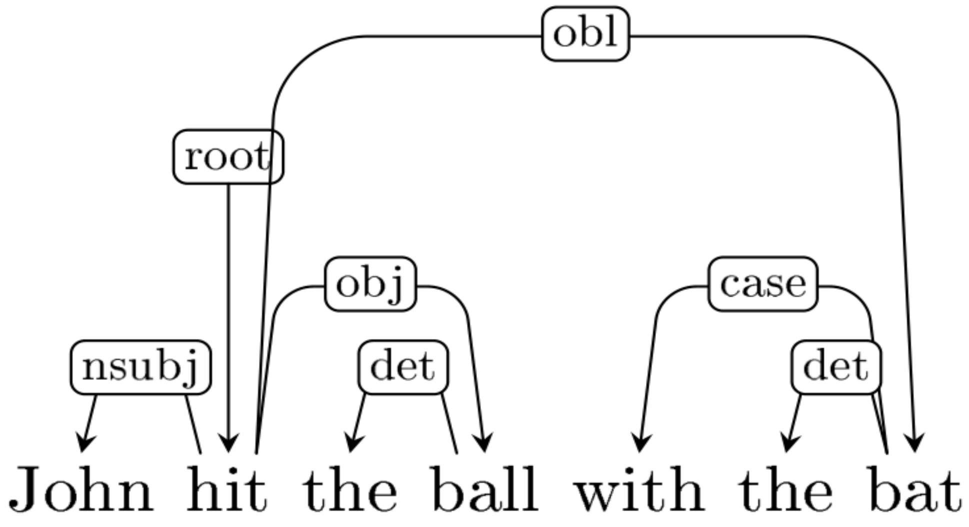
$$score(i,j) = \theta_2^t \cdot \phi_2(i,j,x_{1:n})$$

- Inference (finding the highest scoring tree given θ) is NP-complete. However, there are approximate algorithms.

Online Learning of Approximate Dependency Parsing Algorithms, McDonald and Pereira, EACL 2006

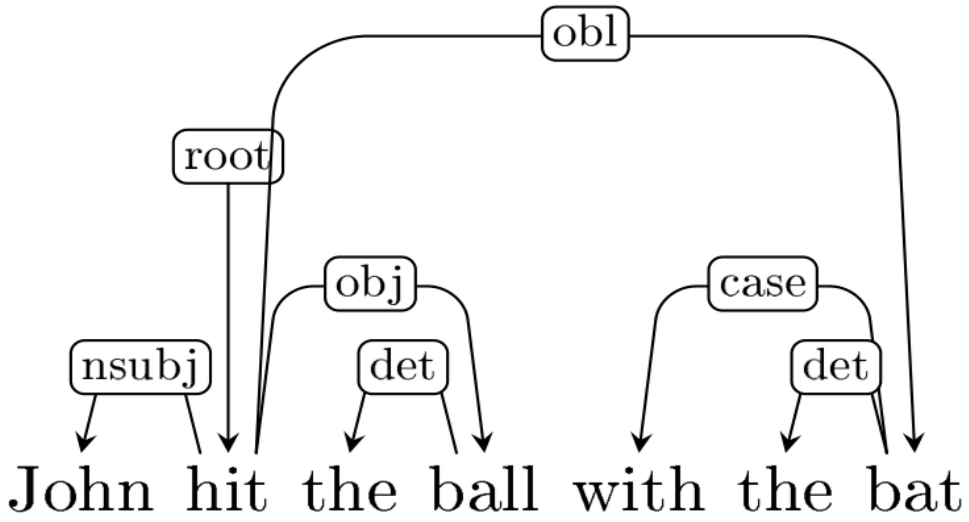
Graph Based Parsing – Higher Order Features

- **Sibling features** are features defined over two edges that have the same parent
- They can be helpful, because once you know "hit → ball" is an edge, "hit → bat" becomes more likely (over "ball → bat")



Graph Based Parsing – Higher Order Features

- **Grandparent features** are features defined over two edges that form a chain
- They can be helpful, because "hit → bat" becomes much more likely once "bat → with" is included



Valence Features

- Verbs tend to appear in specific patterns, in terms of their number of arguments, their order and their prepositions
  - Subject gave Object_1 Obect_2
  - Subject gave Obect_2 to Object_1
  - Object_1 was given to Object_2
  - It was Object_2 that was given to Object_1
  - …
- Encoding what sub-categorization frame is associated with each verb can promote subcategorization frames that appeared during training
- This feature is defined over all children of the verb and some of their children (higher than second-order)

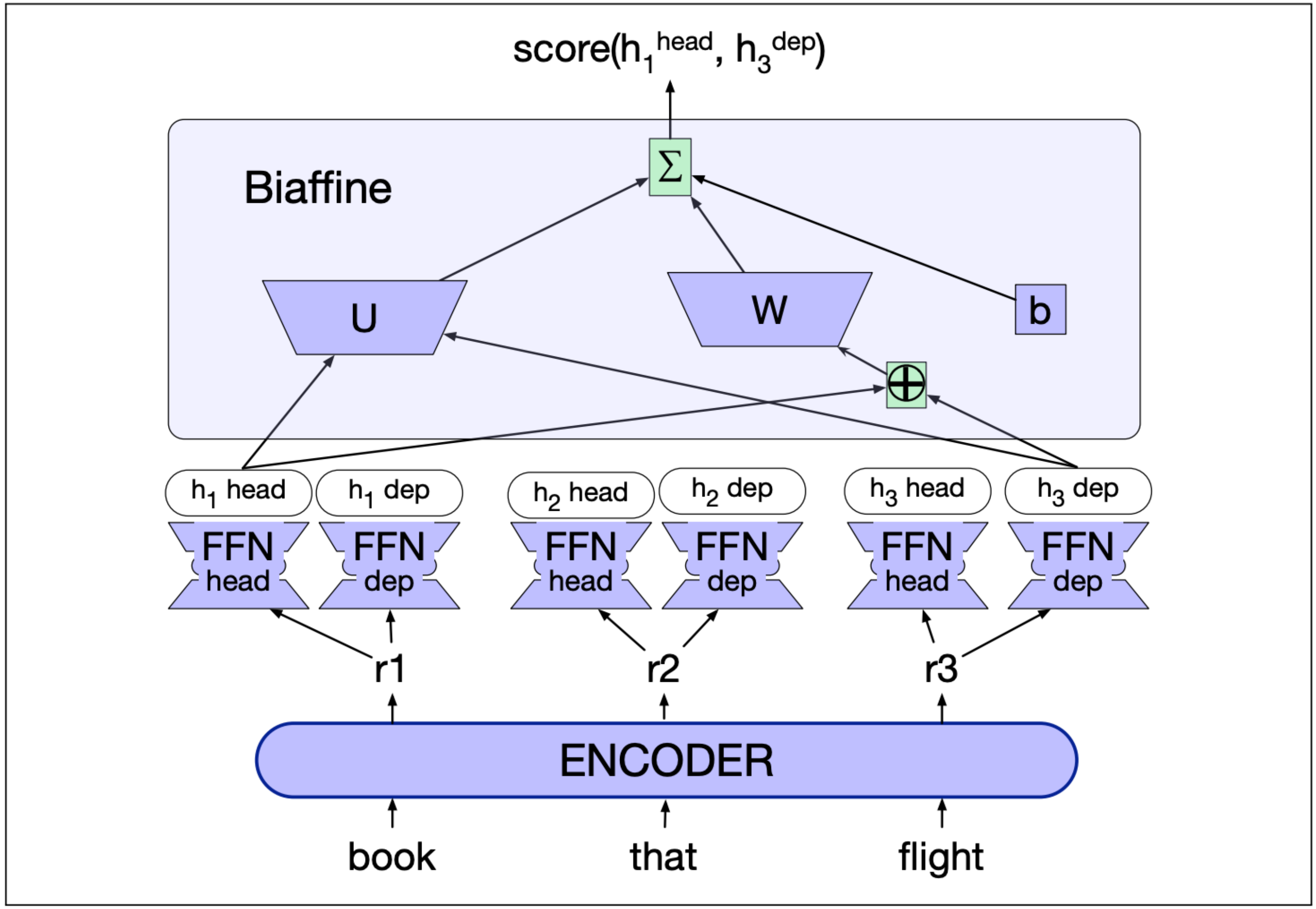
Graph-based Parsing – Higher Order Features

- A lot of interesting machine learning research has been done on building learning models for higher-order features

- A few examples:
  - Xavier Carreras, 2007. Experiments with a Higher-Order Projective Dependency Parser. In CoNLL
  - Yuan Zhang, Tao Lei, Regina Barzilay, and Tommi Jaakkola. 2014. Greed is good if randomized: New inference for dependency parsing. In EMNLP
  - Ilan Tchernowitz, Liron Yedidsion and Roi Reichart. 2016. Effective Greedy Inference for Graph-based Non-Projective Dependency Parsing. In EMNLP

Neural MST Parsing

- Dozat and Manning (2017) proposed using MST only during inference, and to train edge scores using neural networks
- The words are first encoded using a bi-RNN encoder, which produces a word embedding for each word $r_1,\ldots,r_n$
- Each embedding is fed into two networks, producing two series $h_i^{(head)}$ and $h_i^{(dep)}$



Deep Biaffine Attention for Neural Dependency Parsing. Dozat and Manning, ICLR 2017

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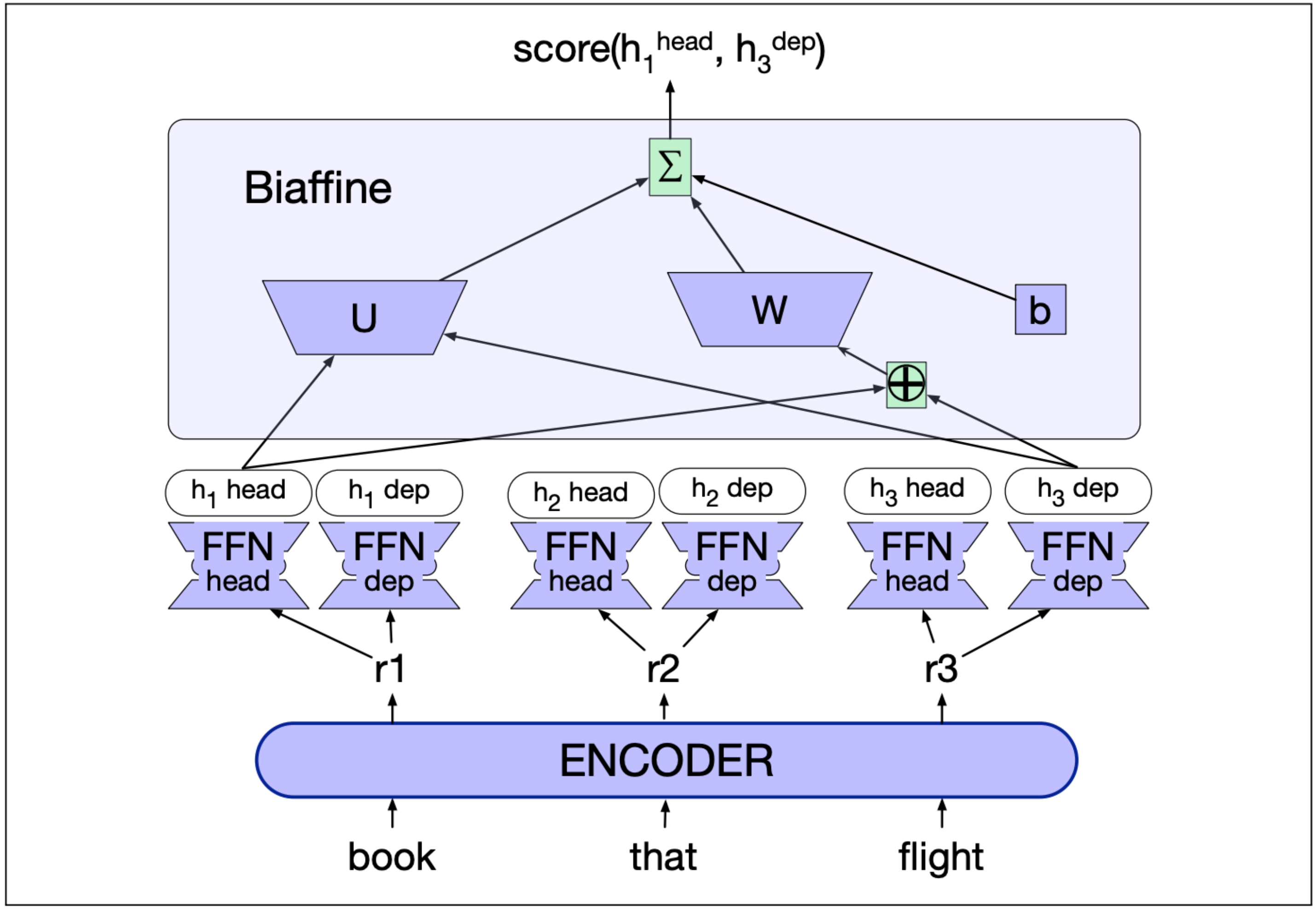

Deep Biaffine Attention for Neural Dependency Parsing
Dozat and Manning, ICLR 2017

Neural MST Parsing: Unlabeled Variant

- We then define a distribution for each word *j*, a distribution over its potential headwords *i*

$$
\begin{aligned}
\mathrm{Score}(i \rightarrow j) &= \mathrm{Biaff}(\mathbf{h}_i^{head}, \mathbf{h}_j^{dep}) \\
\mathrm{Biaff}(\mathbf{x}, \mathbf{y}) &= \mathbf{x}^\top \mathbf{U} \mathbf{y} + \mathbf{W}(\mathbf{x} \oplus \mathbf{y}) + b
\end{aligned}
$$

$$p(i \rightarrow j) = \mathrm{softmax}([\mathrm{Score}(k \rightarrow j); \forall k \neq j, 1 \leq k \leq n])$$

- Training is done locally (i.e., the prediction of a word's head is viewed as independent from the prediction of other words' heads.

Neural MST Parsing: Labeled Variant

- Inference is done through standard MST inference

- For labeled parsing, Dozat and Manning train two classifiers: the one seen last slide, and another label-scorer, which has the same structure, but whose prediction is:
  - Given a dependency edge, predict a softmax over the possible dependency label
  - Inference: first predict the unlabeled dependency, then given the predicted edge, predicts its most likely label

Some Results

- The basic MST parser scores 90.9% LAS on English (Wall Street Journal articles, in domain setting)
- Koo & Collins 2010, third-order dependency parser: ~93% UAS on the same domain
- Dozat & Manning 2017, deep biaffine parser: 95.7 UAS / 94.1 LAS.

- Small, albeit consistent gains, from using higher-order features with neural representation

https://aclanthology.org/2020.acl-main.776.pdf